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BSHES740 2026: Assignment 1 Report

Introduction. The COVID-19 pandemic disproportionately affected the health and wellbeing of people who use opioids (PWUOs). Early in the pandemic, social distancing and other necessary coordinated efforts threatened disruption of care for PWUOs (Alexander, Stoller, Haffajee, & Saloner, 2020). For example, patients receiving methadone are typically required to present daily for directly observed daily dosing, which greatly increased the risk of COVID-19 infection in 2020 (Cowan, Khan, Shastry, & Edelman, 2021). In addition, the added difficulty of accessing illegal drugs during the pandemic was expected to increase production of more powerful substances, catalyzing the shift in demand from heroin to fentanyl (Wakeman, Green, & Rich, 2020). However, no real-time data could be collected to monitor these concerning trends. In this work, we use Natural Language Processing (NLP) techniques to analyze discussion on Twitter about fentanyl, methadone, and Suboxone. Using English-language samples of tweets from just before the COVID-19 pandemic began and at the onset of the pandemic, we find that fentanyl was the most common of these three, but discussion of methadone clinics increased in frequency at the onset of the pandemic.

Methods. One sample of tweets ("Set 1") was collected from shortly before the COVID-19 pandemic began (November 1, 2019 – January 30, 2020) and a second sample ("Set 2") was collected from the beginning of the pandemic period (February 1 – April 29, 2020). For each tweet, the date, tweet content, and Twitter-assigned ID and language were collected. Tweets labelled as non-English were then excluded to avoid confusion; the English term "fent" used as a slang abbreviation for "fentanyl" was a topic of interest, but the same word "fent" is a common helper verb in other languages including Spanish, French, and Catalan. Tweet content was preprocessed, converted to a Corpus, and transformed to two document-term matrices (using term frequency and binary term weighting) using the R packages *stringr* and *tm*. All varieties of whitespace were converted to a space to allow separating words most accurately. We then stripped whitespace, transformed text to lower, handled punctuation and symbols, removed numbers, removed URLs, removed common stopwords, and stemmed all words. In handling symbols, all underscores were removed, since these are commonly used within Twitter handles that are often mentioned within the text of a tweet. However, for all other punctuation and symbols, the character was replaced with a space to most accurately interpret written lists as discrete words; many users opt to write a series of topics in a tweet without using spaces, as in the examples "Meth,Heroin,Fentanyl" and "US+EU+Australia+etc."

Terms matching the strings "fent," "metha," "doloph," "disket," and "subox" were manually reviewed for relevance; "doloph" and "disket" were added because "dolophine" and "diskets" are common brand names for methadone (Methadose (Oral), 2025). Terms that were clearly unrelated were excluded, such as the brand names "Fentiman's" and "Fenty Beauty" or Twitter handles such as "gracefentiman" or "fentmaryvonn." The frequencies of each specific term "fentanyl," "methadone," and "suboxone" were calculated, as well as the cumulative frequency of appearances of all remaining terms matching the related strings above. Subsequent analysis focused on the specific words, since almost all other related terms were relatively uncommon. Topics most closely associated with each of the words "fentanyl," "methadone," and "suboxone" were identified based on correlation between term frequency vectors across each of Set 1 and Set 2. Word clouds were created using the R package *wordcloud* to more clearly visualize common terms across the two datasets. The most common bigrams appearing across the two datasets were analyzed using the R package *ngramrr*. A total of 330 tweets included only one word after preprocessing and were excluded from this analysis (126 from Set 1 and 204 from Set 2).

Results. Set 1 included 171,012 tweets with 66,592 labelled as English; Set 2 included 226,851 tweets with 62,132 labelled as English. Of these 128,724 total tweets, 92,032 contained a mention of "fentanyl" or a related term; 20,957 contained a mention of "methadone" or a related term; and 9,436 contained a mention of "Suboxone" or a related term. More specifically, 81,816 tweets contained the word

"fentanyl" (88.9% of all related terms' tweets), 18,082 contained the word "methadone" (86.28% of all related terms' tweets), and 9,131 tweets contained the word "suboxone" (96.77% of all related terms' tweets). The most commonly discussed analog of fentanyl was carfentanil, appearing in 1,208 tweets (1.31% of all fentanyl-related terms' tweets). After this, the next most common analogs of fentanyl discussed were sufentanil, alfentanil, and remifentanil (28, 26, and 26 tweets). Comparing Set 1 to Set 2, fentanyl appeared 0.77 times per tweet before the pandemic and 0.59 times per tweet during the pandemic; methadone appeared 0.13 times per tweet before the pandemic and 0.17 times per tweet during the pandemic; and Suboxone appeared 0.09 times per tweet before the pandemic and 0.06 times per tweet during the pandemic (see Figure 1). Other common words appearing in Set 1 included "drug," "opioid," "like," "amp," "use," and "people" (see Figure 2); additional words that became more common in Set 2 included "get" and "just" (see Figure 3).

Topics closely associated with fentanyl referred to: danger, such as "illicit," "kill," "overdose," "death," and "lace" (r = 0.16, 0.14, 0.14, 0.13, 0.12); other drugs, such as "heroin," "drug," and "cocaine" (r = 0.14, 0.14, 0.11); and also "China" (r = 0.15). Topics closely associated with methadone referred to: receiving treatment, such as "clinic," "treatment," "program," "daily," "access," and "patient" (r = 0.33, 0.19, 0.10, 0.09, 0.08, 0.08); alternative regimens, such as "buprenorphine" and "withdraw" (r = 0.14, 0.09); and also the label "addict" (r = 0.13). Topics most closely associated with Suboxone overlapped with those associated with methadone, including "addict," "buprenorphine," and "bupe" (r = 0.17, 0.13, 0.12), but also included the term "pain" (r = 0.14) and included several Twitter handles such as "danlairdmd," "fightpaindaily," "process_x," and "ravensspirit" (r = 0.16, 0.14, 0.13, 0.12); this may indicate that discussion of Suboxone in our sample of tweets was dominated by a small number of threads, and we are capturing the names of the users participating in that thread.

The frequency of pairs of words in sequence ("bigrams") was also reviewed for each set. Eight of the ten most common bigrams in Set 1 were also included in the ten most common bigrams in Set 2; but the phrases "fentanyl patch" and "meth fentanyl" became more common than "fentanyl lace" and "lace fentanyl" during the pandemic (see Figure 4). The phrase "methadone clinic" had the sharpest increase in prevalence, from 0.016 to 0.024 appearances per tweet. In both samples, "U S" was one of the most common two-word phrases, suggesting that these English-language samples were primarily focused on United States treatment and policies.

Discussion. This work shows that in our sample of tweets, fentanyl is much more commonly discussed than methadone and methadone is slightly more commonly discussed than Suboxone. We find that discussion of methadone increased during the pandemic, and methadone clinics were particularly discussed more commonly. This finding confirms experts' initial concerns about the pandemic's impact on PWUOs: the daily regimen of monitored, in-person methadone treatment became much more complicated amid social distancing at the onset of the pandemic, likely generating many related topics of conversation. For fentanyl, the substance's danger explains several closely related topics, but the fact that "China" was also a closely associated topic may indicate that some of the discussion was politicized, particularly since 2020 was a year leading up to a presidential election. This increases the difficulty of using Twitter to understand the specific experiences of PWUOs during the pandemic. Lastly, we note that this analysis was limited to the English-language tweets in our original sample; this is a limitation, and it would likely be useful for other researchers to use these tweets to understand discussions around fentanyl, methadone, and Suboxone in other languages used in the United States and in other countries around the world.

Works Cited

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Methadose (Oral). (2025, October 10). Retrieved from Drugs.com: https://www.drugs.com/cons/methadose.html#brand-names

Wakeman, S. E., Green, T. C., & Rich, J. (2020). An overdose surge will compound the COVID-19 pandemic if urgent action is not taken. *Nature Medicine, 26*, 819-820.

Supplemental Materials

Code and tables associated with this report can be found in the public GitHub repository: https://github.com/ryantaylor1997/bshes740_assignment1_taylor.

Appendix

See figures below.

Figure 1

Appearances per Tweet, Pre-COVID and During COVID

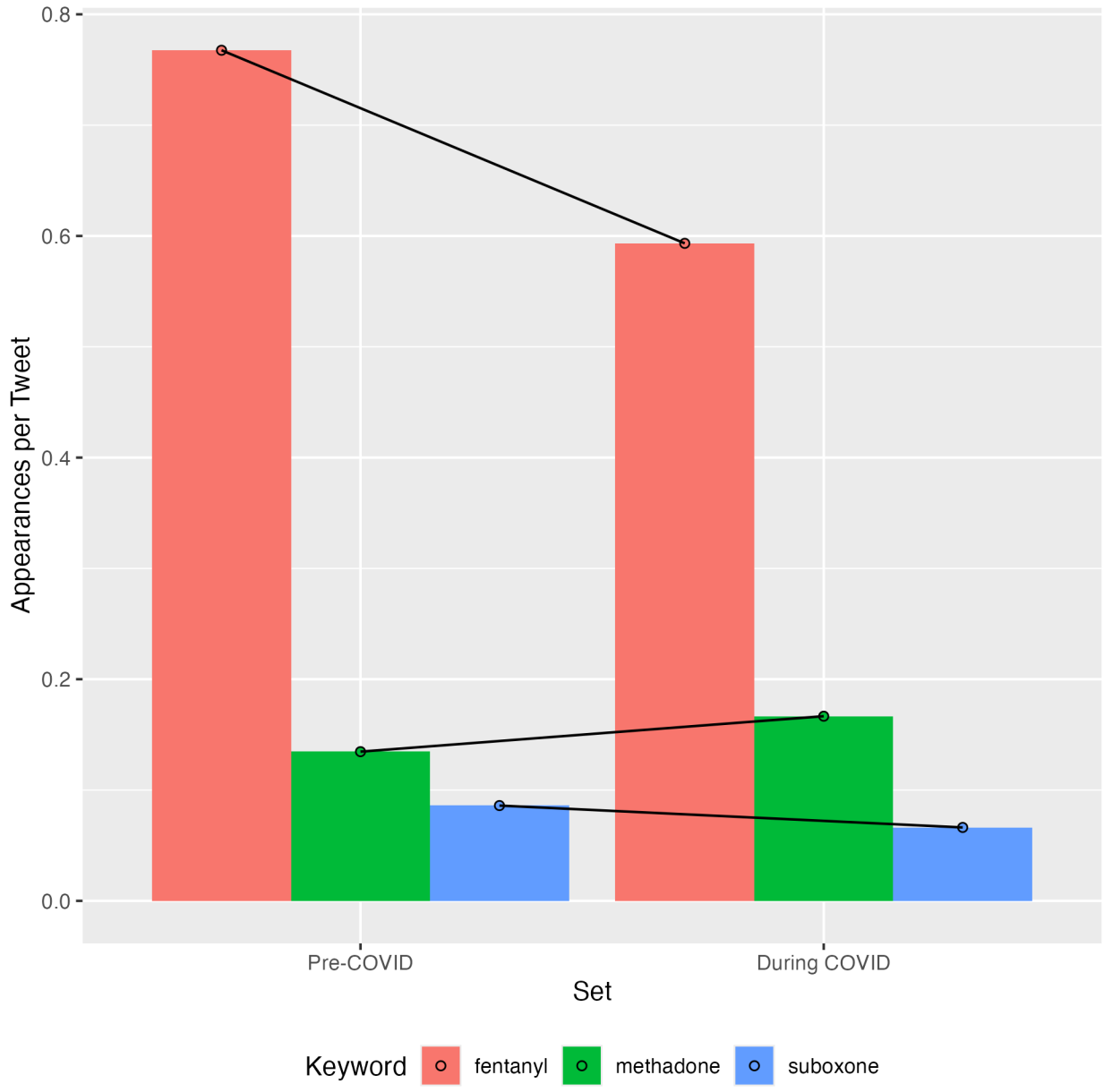


Figure 2

Common terms in Set 1



Figure 3

Common terms in Set 2

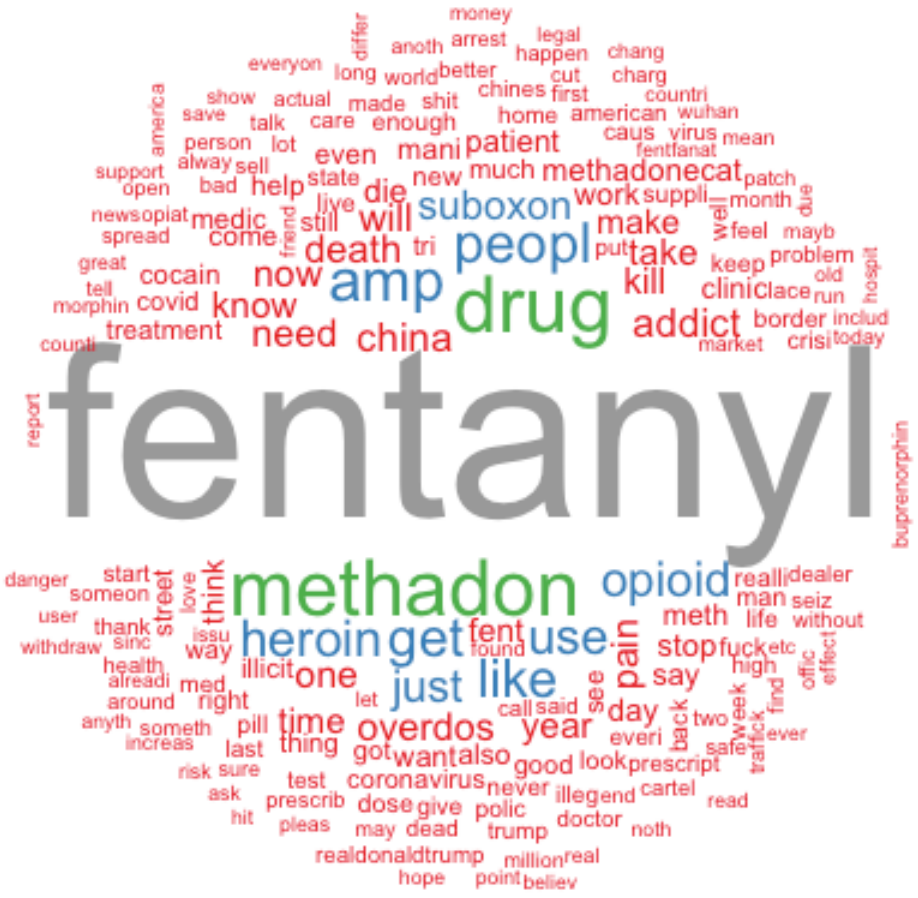


Figure 4

Most common bigrams in Set 1 and Set 2

